Carbohydrates and the human gut microbiota

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Purpose of review
Due to its scale and its important role in maintaining health, the gut microbiota can be considered as a 'new organ' inside the human body. Many complex carbohydrates are degraded and fermented by the human gut microbiota in the large intestine to both yield basic energy salvage and impact gut health through produced metabolites.

Recent findings
This review will focus on the gut microbes and microbial mechanisms responsible for polysaccharides degradation and fermentation in the large intestine. Gut microbes and bacterial metabolites impact the host at many levels, including modulation of inflammation, and glucose and lipid metabolisms.

Summary
A complex relationship occurs in the intestine between the human gut microbiota, diet and the host. Research on carbohydrates and gut microbiota composition and functionality is fast developing and will open opportunities for prevention and treatment of obesity, diabetes and other related metabolic disorders through manipulation of the gut ecosystem.

Keywords
carbohydrates, gut microbiota, intestinal fermentation, short chain fatty acids

INTRODUCTION

The human gut microbiota is a dense and diverse microbial community of billions of microbes. Most of them are strict anaerobes and colonize the human large intestine which is a highly specialized organ acting as an anaerobic digester of complex molecules including carbohydrates. The gut microbiota helps extract nutrients and energy from the diet [1▪▪], but also plays a major role in health and disease, protecting the host from digestive infections and contribute to the maturation of the immune function [2,3]. Microbial perturbations and even characterized dysbiosis of the human gut microbiota have been reported for many diseases and disorders including diabetes [4], malnutrition [5] and obesity [6]. The human gut microbiota composition and metabolic activities are impacted by the diet and reciprocally, host metabolism and metabolites also interact with the gut microbiota and diet shaping a complex interaction network driving gut health (Fig. 1). It is, therefore, crucial to better understand the relationships between host, diet and the gut microbiota, starting by investigating the specific effects of the main components of the diet.

This review will focus on recent findings of the impact of carbohydrates (unabsorbed sugars, oligosaccharides and complex polysaccharides) on the colonic microbiota and human host, highlighting potential effects for human health.

THE HUMAN GUT MICROBIOTA

The human gut microbiota is a dense and diverse bacterial community colonizing the different parts of the gastrointestinal tract but massively located in the large intestine (making up to 10^{14} bacterial cells). Taxonomic composition of the human gut microbiota has been first studied with culture-dependent methods and more recently culture-independent molecular methods, from fluorescence in situ hybridization (FISH) analysis to microarray and high throughput sequencing used for in-depth analysis of the intestinal microbial community [7–9].

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KEY POINTS

- The human gut microbiota is a complex microbial community of billions of microbes benefiting the host by extracting nutrient and energy from complex polysaccharides but also playing a major role in health and disease.
- Humans consume dozens of polysaccharides that are not degraded by host enzymes and are fermented by the colonic gut microbiota.
- Characterization of gut microbes involved in polysaccharides fermentation is important to understand bacterial mechanisms and further understand the impact of these microbes and their metabolites on the host.
- Better comprehension of interactions occurring in the human gut between the microbiota, diet and the host will open the door to new nutritional approaches and treatments.

The human gut microbiota is mainly dominated by the two major phyla, Bacteroidetes and Firmicutes, whereas Actinobacteria, Proteobacteria and Verrucomicrobia are found at lower levels [10] (Fig. 2). The colonic microbiota consists of several hundred species belonging to these five phylogenic groups, but the composition of the human microbiota at the species level is a unique combination of microorganisms. The colonic microbiota is usually dominated by *Bacteroides* and *Prevotella* species (Bacteroidetes), *Clostridium, Roseburia, Faecalibacterium* and *Ruminococcus* species (Firmicutes), *Bifidobacterium* species (Actinobacteria) or *Akkermansia* sp. (Verrucomicrobia) (Fig. 2). *Archaea methanogens* can be present at high levels in methane-excreting individuals and inexplicably at very low level (sometimes even nondetectable) in nonmethane-excreting individuals [11].

The human gut microbiota is considered to be a very stable and resilient ecosystem. The presence of a core gut microbiota has been suggested [12] while three distinct enterotypes have also been reported [9]. Major functional groups of microbes and population levels appear to be shared between individuals and also to be rather stable [7]. These characteristics have been confirmed recently by metagenomic approaches, extensive functional gene sequencing and pathway characterization [9,13]. Interestingly, the predominant functional pathways found in the gut microbiota are usually linked to carbohydrate fermentation and degradation pathways [5].

SOURCES OF FERMENTABLE CARBOHYDRATES IN THE LARGE INTESTINE

The fraction of the diet reaching the large intestine is largely composed of complex polysaccharides containing bonds that are resistant to host digestive enzymes. A large diversity of complex glycans are then available for the colonic microbiota metabolism including undigestible polysaccharides, but also monosaccharides and disaccharides not fully absorbed in the upper part of the gastrointestinal tract, due to overconsumption or lack of digestion, and some endogenous substrates (glycoproteins; Table 1) [11,14▪,15,16,17,18,19,20,21,22,23,24▪,25,26,27,28].

Resistant starch (RS1 to RS4) is quantitatively one type of the major complex polysaccharides supplied to the microbiota through the diet. Humans secrete enzymes (amylases and glucoamylases) that completely degrade starch but not resistant starch. Moreover, nonstarch polysaccharides (NSPs), including plant cell wall polysaccharides (cellulose, hemicellulose, lignin) and storage polysaccharides (pectin and oligosaccharides), are not degraded in the upper

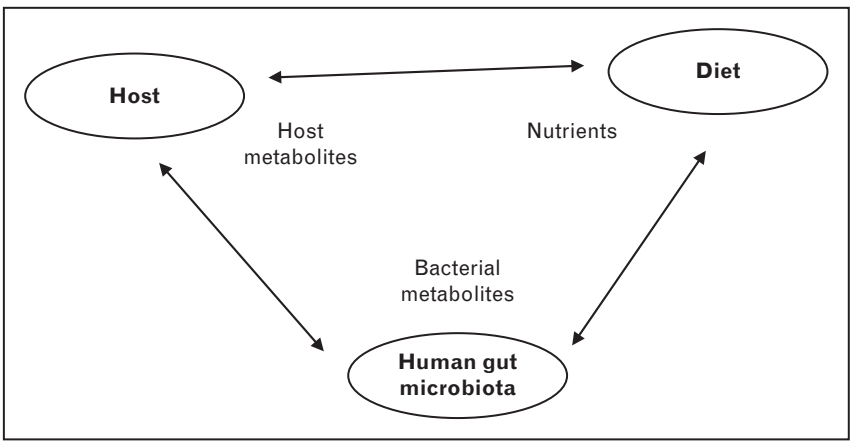


FIGURE 1. Schematic ecological interactions occurring in the human colon.

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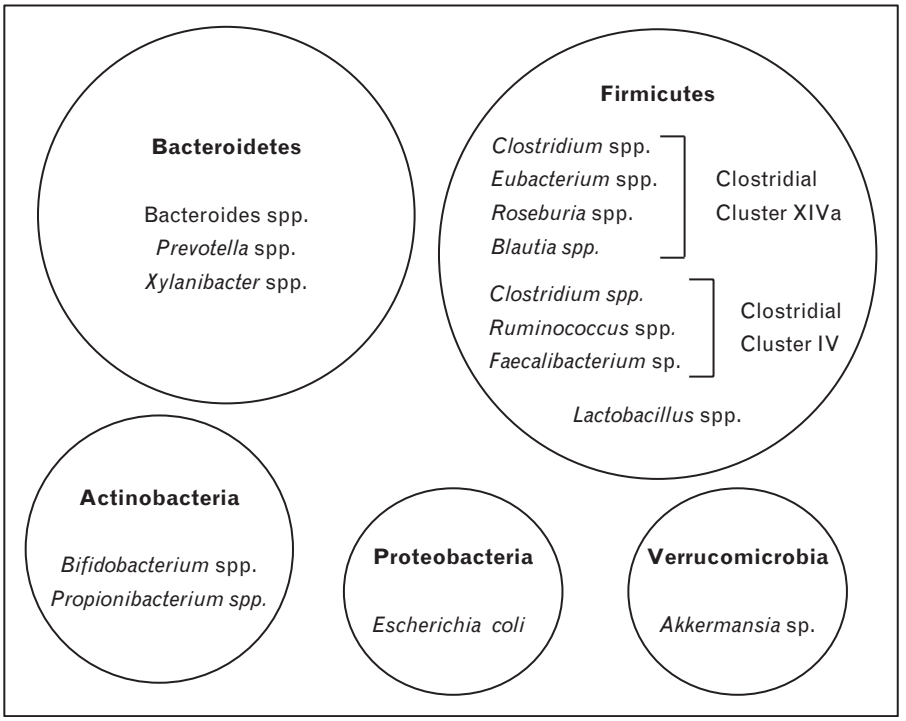


FIGURE 2. Predominant species of the human colonic microbiota.

part of the gastrointestinal tract and can be fermented by the colonic microbiota. In infants, large amounts of diverse human milk oligosaccharides (HMOs) are supplied by the breast milk [21]. Additionally, monosaccharides and disaccharides such as lactose, fructose and sugar alcohols (sorbitol, lactitol and other polyols) can also reach the large intestine when intestinal malabsorption or overfeeding of these sugars, widely used for the formulation of processed foods or beverages, occur [24▪]. Endogenous glycans of host origin such as mucopolysaccharides (glycoproteins) or exopolysaccharides (EPSs) of bacterial origin are also available to the human gut microbiota.

THE HUMAN GUT MICROBIOTA FERMENTS SUGARS AND COMPLEX POLYSACCHARIDES

In the large intestine, complex polysaccharides are extensively degraded and fermented by the gut microbiota, except lignin that is not digested

Table 1. Overview of carbohydrates available for the colonic gut microbiota and their major bacterial utilizers

Substrates	Origin	Major bacterial utilizers	References
Resistant starch (RS1 to RS4)	Dietary fiber	*Ruminococcus, Bacteroides*	[14▪]
Hemicelluloses (xylan and arabinoxylan)	Dietary fiber	*Roseburia, Bacteroides, Prevotella*	[15,16]
Cellulose	Dietary fiber	*Ruminococcus, Bacteroides*	[11]
Pectin	Dietary fiber	*Bacteroides, Faecalibacterium*	[17,18]
Fructans (inulin and FOS)	Dietary fiber and oligosaccharides	*Bacteroides, Roseburia, Faecalibacterium, Bifidobacterium*	[19,20]
Milk oligosaccharides	Dietary oligosaccharides	*Bifidobacterium*	[21]
Lactose (if malabsorption)	Dietary sugar	*Lactobacillus, Bifidobacterium*	[22]
Fructose (if malabsorption and overfeeding)	Dietary sugar	*Bifidobacterium, Roseburia*	[23,24▪]
Sugar-alcohols (sorbitol, mannitol and so on)	Dietary sugar-alcohols	*Lactobacillus, Escherichia*	[24▪,25]
Mucins and mucopolysaccharides	Endoglycoproteins	*Akkermansia, Bacteroides*	[26,27]
Bacterial polysaccharides	Endogenous bacterial polysaccharides	*Bifidobacterium, Anaerostipes, Prevotella*	[28]

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and cellulose that is poorly fermented in certain individuals [11]. The fibrolytic community degradation is primarily responsible for complex fiber fermentation and involve specific anaerobic gut microbes producing enzymes. They belong to the predominant bacterial groups including dominant *Bacteroides* (resistant starch, xylan), *Roseburia* (resistant starch, xylan and oligosaccharides), *Ruminococcus* (resistant starch, cellulose) or *Bifidobacterium* (oligosaccharides) species (Table 1). It can be mentioned that a single microbe can exhibit several fibrolytic activities, contributing to the degradation of several complex polysaccharides. Key fibrolytic species have been identified but seem to be especially effective to ferment a certain type of individual polysaccharides. For example, *Ruminococcus bromii* is one of the major starch degraders of the human colon [14▪], whereas *Ruminococcus champanellensis* or *Bacteroides xylanisolvens* significantly contribute to the degradation of cellulose and xylan, respectively [29,30].

Complex fiber degradation is nevertheless a complex process involving a large number of different fibrolytic species in a bacterial consortium in order to circumvent accessibility issues of the substrate and degrade highly complex and resistant vegetal structure (especially plant cell wall polysaccharides). Mucopolysaccharides (endogenous glycoproteins) are degraded by the predominant *Bacteroides* species, but also by a specialized microorganism, *Akkermansia muciniphila* [31] (Table 1). These fibrolytic species are recognized as primary degraders and further release sugars that can be used for their own needs but also by glycolytic bacteria unable to degrade complex fibers. This phenomenon called crossfeeding characterizes nutritional interactions in bacterial communities organized in a trophic chain (Fig. 3). Bacterial metabolites produced by the fibrolytic and glycolytic bacteria are continuously used by other groups of microbes which produce other metabolites further used by new microbes until an equilibrium is reached allowing optimized fiber degradation and energy salvage.

Among metabolites, H_2 is one of the key regulators of organic matter degradation. H_2 is largely produced during fiber and sugar fermentation and it has been shown that its accumulation can inhibit further organic matter oxidation in a negative feedback process depending on the partial pressure of H_2 in the gut [32]. H_2 is, therefore, mainly dissipated by hydrogenotrophic microbes, converting H_2 into other metabolites such as methane (methanogens), acetate (hydrogenotrophic acetogens) or acetate and H_2S (sulphate-reducing bacteria) [7]. Therefore, H_2 metabolism by the gut microbiota is a central process in order to achieve effective gut fermentation (Fig. 3).

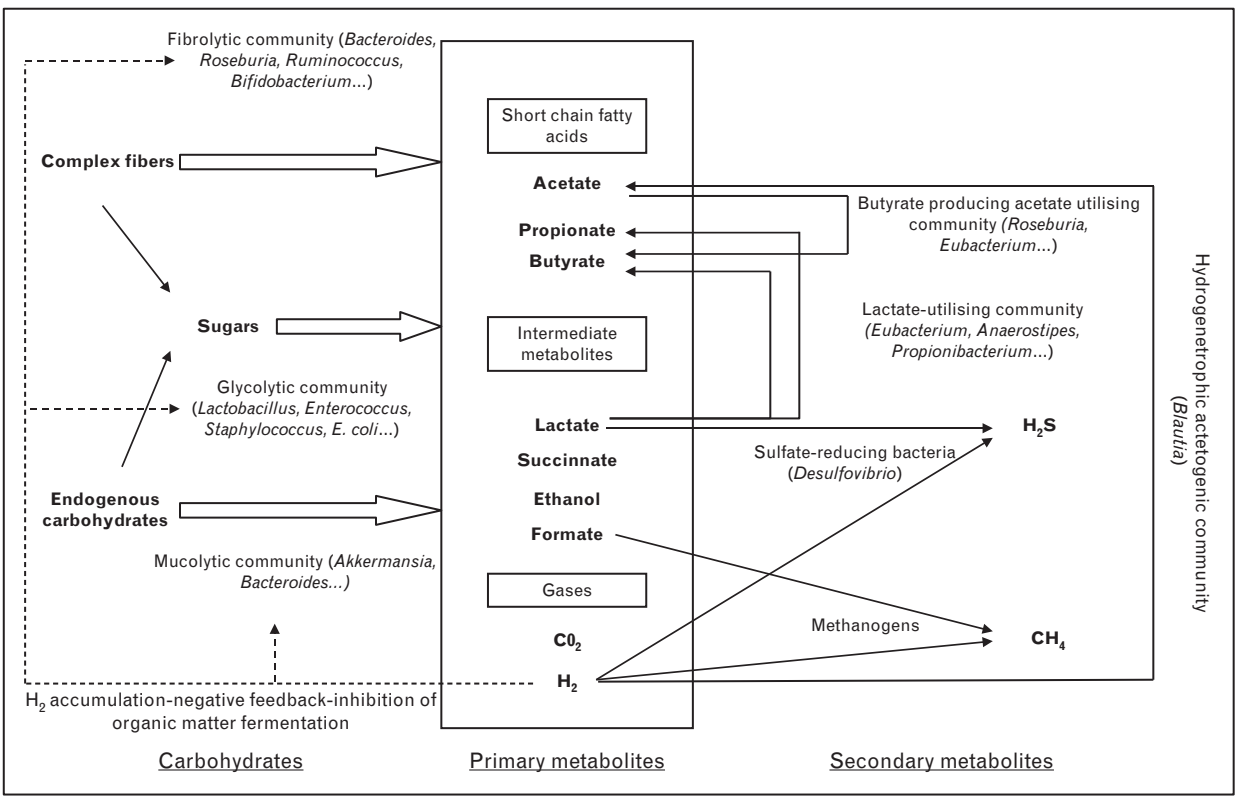


FIGURE 3. Fermentation of carbohydrates by functional group of microbes of the human colonic microbiota.

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Fructose can be taken up in the small intestine but is also one of the most poorly absorbed short-chained carbohydrates, with 50 g free fructose capable of inducing malabsorption in 80% of people [24▪]. Lactic acid bacteria (mainly *Lactobacillus* species) are predominant fructose utilizers, but this activity has been reported in Clostridium cluster IV genus *Faecalibacterium* [33]. Unabsorbed sugar alcohols (e.g. sorbitol, mannitol, glycerol) commonly used in foods as bulking compounds and sugar replacers and also to lower water activity for food preservation, can be fermented by lactic acid bacteria but also by γ-Proteobacteria species including *Escherichia coli* and *Salmonella* species [24▪].

CARBOHYDRATES SHAPE THE HUMAN GUT MICROBIOTA

Carbohydrate availability in the large intestine can shape the whole microbial community by impacting the fibrolytic communities [34], but also all other functional groups of microbes connected in cascade to these primary degraders [32]. Recent studies have investigated the human gut microbiota of individuals from different geographical origins but also with very different diet habits. The gut microbiome of rural African infants fed with plant-based diet is colonized by the predominant *Prevotella* and *Xylanibacter* species that were not found in urban European infants in the same study [16]. Major differences in bacterial populations at the phylum level have even been reported for these two groups of infants, with more Bacteroidetes than Firmicutes in African infants in contrast to European infants showing an inverse ratio. A recent study comparing the gut microbiome of adults from different continents (Africa, North and South America) confirmed large variations among Americans, Europeans and Africans. The gut microbiomes of Africans clustered with South Americans correlating with diets high in plant-based polysaccharides but were different from North Americans with low fiber consumption. Carbohydrates in the diet could, therefore, contribute to the large variation observed for gut microbiota composition [1▪▪].

The effect of polysaccharides has been also investigated in human dietary intervention studies. The use of depleted or enriched diets in polysaccharides revealed that resistant starch or fructans can shape the gut microbiota. Among studied bacterial groups, *Ruminococcus*, *Roseburia* or *Bifidobacterium* strongly responded to dietary changes. A decrease in total carbohydrate intake in obese adults led to a decrease in *Roseburia-Eubacterium rectale* species [35], whereas the supplementation of the diet with resistant starch increased *Ruminococcus* (*R. bromii* and relatives) and *Eubacterium rectale* [36,37], or *Bifidobacterium adolescentis* and *Parabacteroides distasonis* according to the substrate considered [37]. Fructans, inulin and fructooligosaccharides (FOSs) are also well known to specifically change the gut microbiota composition. Many intervention studies in humans have demonstrated that inulin and FOSs promote the growth of 'healthy bacteria', including *Bifidobacterium* species and sometimes *Lactobacillus* species, whereas *Bacteroides* and *Clostridium* were decreased with fructan supplementation. Fructans can be potentially degraded by many other microbes including *Roseburia* species [38]. Recent studies confirmed that fructans can especially promote the growth of butyrate-producing bacteria such as *Faecalibacterium prausnitzii* [20,39]. Prebiotics are generally recognized to enhance the barrier effect of the gut microbiota toward enteropathogens, but few studies with animal models have shown that prebiotics, especially inulin and FOSs, could promote *Salmonella* growth, translocation or virulence [40–43]. Such adverse effects have been also observed *in vitro* using a novel *Salmonella* intestinal fermentation model [44]. Therefore, the safety to use prebiotics in pathogen-contaminated environments should be investigated further.

Carbohydrates can also shape the metabolic activities of the human gut microbiota. Fermentation of carbohydrates by the gut microbiota leads to the production of short chain fatty acids (SCFAs) and gases (CO_2 and H_2). A recent in-vitro fermentation study has investigated the impact of high energy, normal energy and low energy media on the gut microbiota, showing that a high energy fructose enriched medium was strongly butyrogenic [45]. Fructans are also known to promote butyrate production in the human gut by stimulating the growth of butyrate-producing bacteria or through bacterial crossfeeding [46,47]. Fructose and sugar alcohols can be fermented by lactic acid bacteria to produce lactate further reused by lactate-utilizing butyrate-producing bacteria to metabolize butyrate and then contribute to the intestinal energy metabolism [48]. However, increased dietary energy extraction and metabolism may promote metabolic disorders [24▪]. Many other bacterial metabolites produced by the human gut microbiota can also impact gut health.

HOST EFFECT OF SUGARS AND COMPLEX POLYSACCHARIDE CARBOHYDRATE FERMENTATION

Complex carbohydrates and sugars are metabolized into SCFAs, acetate, butyrate and propionate, by the colonic microbiota. This process also affects

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the gut environment, for example by decreasing pH, as large amounts of acids are produced. Low pH can locally impact the growth of enteropathogens but also modify metabolite absorption of the colonocytes. SCFAs can be used as an energy source for the host. Butyrate is locally metabolized by colonocytes, whereas acetate and propionate are carried in the bloodstream to the liver for gluconeogenesis (propionate) and to the heart or brain for oxidation (acetate) [21].

SCFAs are also signaling molecules impacting gut motility, in addition to regulation of transit time by fibers, and epithelial cell proliferation [49]. SCFAs trigger secretion of glucagon-like peptide (GLP)-1 in cell models [50▪▪] and impact olfactory receptors modulating blood pressure [51]. Acetate can induce the epithelial expression of the antimicrobial peptide LL37 [52] or sense GPR41 and GPR43 on intestinal immune cells and it regulates inflammation [53]. In-vitro cellular investigations also demonstrated that butyrate modulates inflammation, immune cell function and migration [54]. This effect of butyrate could explain the benefits of resistant starch and prebiotics (fructans and galactooligosaccharides, GOSs) in animal models [55,56] by reducing the severity of induced colitis. The impact of FOS in active Crohn's disease was investigated in a randomized, double-blind, placebo-controlled trial, but this study failed to show clinical benefits even if the immune system was modulated [57]. The effect of GOS was also recently investigated in overweight adults [58]. This study confirmed that GOSs can modulate the immune response in humans. GOS supplementation impacted the gut microbiota composition by increasing *Bifidobacterium* populations but also significantly increased in fecal secretory IgA and decreased in fecal calprotectin and plasma C-reactive protein. In the same experiment, GOS supplementation also reduced markers of the metabolic syndrome such as blood insulin, total cholesterol and the ratio total cholesterol/high-density lipoprotein [58].

The potential of fibers for modulating the metabolic syndrome or diabetes is currently under evaluation and represents an important area of research for gut health. Most studies have, however, been only performed with animal models. Prebiotic enriched diet improved glucose homeostasis, leptin sensitivity in obese and diabetic mice [59]. Resistant starch also enhanced maternal glycemic control in diabetic rats [60], whereas chitin-glucan decreased body weight gain, hyperglycemia, hypercholesterolemia and glucose intolerance in a mice model fed with a high-fat diet [61▪]. These positive results were associated with major changes in the gut microbiota and especially the clostridial cluster XIVa (*Roseburia* spp.). A recent trial performed in obese women failed, however, to see major changes in host metabolism with fructan supplementation even if the gut microbiota composition of the volunteers was clearly impacted [39].

Prebiotics could also impact on satiety. Arabinoxylan oligosaccharides (AXOSs) increase satietogenic gut peptides in a diet-induced obese mice model [62]. In contrast, alginate from seaweeds had no impact on satiation and gut satiety hormones in overweight and obese individuals [63]. Complex carbohydrates from seaweeds can, however, represent a potential source of fibers with prebiotic properties. In-vitro fermentation experiments have recently shown that agar and alginate fermentations can promote SCFA production and bifidobacterial populations in a batch fermentation model [64]. The specific carbohydrate active enzymes of seaweeds polymers needed for intestinal degradation are likely present in low amount and activity in bacteria of the human gut, potentially limiting the fermentation of these polysaccharides. The human gut microbiota can only acquire these enzymatic properties from marine bacteria by the horizontal gene transfer in order to ferment a broader range of new complex polysaccharides potentially beneficial for gut health [65,66]. The combination of seaweeds polymers with probiotic bacteria possessing these enzymatic properties could lead to the development of highly specific new synbiotics for clinical nutrition.

CONCLUSION

Many complex carbohydrates are degraded and fermented by the human gut microbiota in the large intestine. Much more than basic energy salvage, fiber fermentation can impact gut health through produced metabolites. Diet and specific polysaccharides can also change the composition and metabolic activities of the gut microbiota. Supplementation of the diet with specific polysaccharides can promote the growth of healthy microbes (*Bifidobacterium*, *Lactobacillus* or butyrate-producing bacteria), SCFA production and lowering pH, and can therefore lead to inhibitory effects toward pathogens. Fibers and bacterial metabolites can also impact host and gut health by modulating inflammation, and glucose and lipid metabolisms. Additional research is needed to understand the underlying mechanisms and characterize new prebiotic substrates with efficiency in disease prevention or treatment for clinical nutrition. The continuous development of high throughput methods including omics approaches (metabolomics and

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transcriptomics) will also contribute to a better understanding of the interactions between dietary carbohydrates, the gut microbiota and host.

Acknowledgements

None.

Conflicts of interest

There are no conflicts of interest.

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- ■ of special interest
- ■■ of outstanding interest

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